

Mechanistic Interpretability of a Plant Genomic Foundation Model: Data Probes, Causal Interventions, and Motif-Grounded Features

Anonymous Authors

Abstract

Foundation models trained on plant DNA are increasingly used in computational plant biology, yet their internal representations remain opaque. We present a mechanistic interpretability study of Plant-DnaGemma, a 152M-parameter Gemma-based transformer, on annotated plant genomic data: 15 000 TAIR10/Araport11 region-type sequences (exons, introns, 5'/3' UTRs, intergenic), 9 000 Arabidopsis splice-site windows, 22 700 EPDnew transcription start sites, 40 700 promoter windows, and 12 000 windows from six species (Arabidopsis, rice, maize, Brachypodium, tomato, soybean). With 5-fold stratified cross-validation and held-out-chromosome test evaluation, the trained model achieves 64% on 5-way region-type classification and exceeds a 4-mer logistic baseline by 10 pp on region-type and 3 pp on 3-way splice-site classification; on binary TSS and promoter tasks Plant-DnaGemma is effectively tied with a 3-layer CNN trained from scratch, an honest negative-style finding. A Top-K sparse autoencoder trained on layer-7 activations yields 99.7% sparsity with 92% variance explained; of 300 analyzed features, **72 (24%) are significantly enriched for a JASPAR 2024 plant transcription-factor motif** ($q < 0.05$, BH), including BPC, DOF, MYB, NAC, and ERF family binding sites, and 47 of the 72 map to named PlantTFDB families. On a GC-matched 4-species subset, Plant-DnaGemma achieves 68% species classification versus a 26% GC-only baseline (chance 25%), and after controlling for GC content a partial Mantel test shows that representational distances correlate significantly with phylogenetic distance ($r = 0.55$, $p = 0.019$). This work establishes a methodological framework—annotated data, learned baselines, GC-matched controls, and motif-grounded feature interpretation—for mechanistic interpretability of genomic foundation models.

1 Introduction

Foundation models trained on DNA have achieved strong performance across genomic prediction tasks [Dalla-Torre et al., 2023, Nguyen et al., 2024, Ji et al., 2021, Zhou et al., 2024]. In plant biology, models such as AgroNT [Mendoza-Revilla et al., 2024], PlantRNA-FM [Yu et al., 2024], Plant-DnaGemma [Zhang et al., 2024], and Caduceus [Schiff et al., 2024] learn representations from raw nucleotide sequences. The internal mechanisms by which these models represent genomic features remain poorly understood.

Mechanistic interpretability—the reverse-engineering of neural network computations [Elhage et al., 2022, Olah et al., 2020, Conmy et al., 2023]—has yielded insights in NLP: circuits for indirect object identification [Wang et al., 2023], modular arithmetic [Nanda et al., 2023], and factual recall [Meng et al., 2022]. In genomics, interpretability work has been largely limited to attention visualization and gradient attribution [Clauwaert and Waegeman, 2021, Avsec et al., 2021, Yu et al., 2024]. No prior work has applied the full modern interpretability toolkit—probing with learned baselines, causal interventions, sparse autoencoders linked to known biology—to plant genomic foundation models on annotated data.

This work fills that gap. We evaluate Plant-DnaGemma (152M parameters, 12 transformer layers) on six annotated benchmarks built from TAIR10/Araport11, EPDnew, and Ensembl Plants release 58, and compare against learned-from-scratch CNN and LSTM baselines. Our contributions are:

1. **Probing with learned baselines.** 5-fold cross-validated probing on 110 000+ annotated plant genomic sequences, with chromosome-split held-out evaluation. Plant-DnaGemma beats k -mer logistic

regression, a 1-layer CNN, a 3-layer DanQ-style CNN, a bi-LSTM, and a k -mer MLP on multi-class region-type (+6.9 pp over the best baseline) and splice-site (+3.4 pp) tasks; on binary TSS and promoter tasks a task-supervised CNN effectively matches the foundation model (Sec. 4.1, 4.2).

2. **Cross-species analysis with controls.** Scaled to 6 species $\times$ 2000 windows $\times$ 1024 nt, with GC-matched and held-out species controls. Plant-DnaGemma captures non-compositional species information: 68% accuracy on GC-matched 4-species classification (chance 25%, GC-only baseline 26%), and partial Mantel correlation of 0.55 ($p=0.019$) between representational and phylogenetic distance *after* regressing out GC (Sec. 4.4).
3. **Motif-grounded sparse features.** A Top-K sparse autoencoder (expansion $16\times$, $L_0=32$) over layer-7 activations decomposes representations into features of which 24% (72 of 300) are significantly enriched (Mann-Whitney, BH $q<0.05$) for JASPAR 2024 plantae motifs, with 47 mapping to PlantTFDB Arabidopsis TF families (BBR-BPC, NAC, MYB, AT-hook, WRKY, Dof, GRF, ERF, and others; Sec. 4.5).

2 Related Work

Genomic Foundation Models. DNABERT [Ji et al., 2021] adapted BERT-style pretraining for DNA with k -mer tokenization; DNABERT-2 [Zhou et al., 2024] replaced k -mers with BPE tokenization. The Nucleotide Transformer [Dalla-Torre et al., 2023] scaled to 2.5B parameters across multi-species genomes. HyenaDNA [Nguyen et al., 2024] demonstrated long-range modeling with sub-quadratic architectures. Caduceus [Schiff et al., 2024] introduced bidirectional equivariant architectures for DNA. In plants, AgroNT [Mendoza-Revilla et al., 2024] trained a 1B-parameter model on 48 plant genomes, and Plant-DnaGemma [Zhang et al., 2024] adapted Gemma with BPE tokenization for plant DNA.

Mechanistic Interpretability. Key techniques include linear probing [Alain and Bengio, 2017, Belinkov et al., 2017, Hewitt and Manning, 2019], activation patching [Vig et al., 2020, Meng et al., 2022], and sparse autoencoders for feature decomposition [Cunningham et al., 2023, Bricken et al., 2023, Templeton et al., 2024, Gao et al., 2024]. Circuit analyses have revealed induction heads [Olsson et al., 2022], indirect object identification circuits [Wang et al., 2023], and knowledge-storage mechanisms [Geva et al., 2023].

Interpretability in Genomics. Prior work has focused on attention visualization [Clauwaert and Waegeman, 2021] and sequence attribution [Avsec et al., 2021]. Probing classifiers have been applied to protein language models [Vig et al., 2021, Rao et al., 2019, Rives et al., 2021] but not systematically to plant DNA models. Our work applies the full modern interpretability toolkit with proper baselines to a plant DNA model on annotated genomic data.

3 Methods

3.1 Model

We analyze zhangtaolab/plant-dnagemma-BPE, a 152.4M-parameter transformer based on Gemma [Gemma Team, 2024]: 12 layers, 12 attention heads per layer (head dim 256), 768-dimensional hidden states, BPE vocabulary of 8002 plant-genome tokens, 1024-token context window. All experiments ran on a single NVIDIA RTX 2060 (6 GB) with Apple-Silicon MPS fallback.

3.2 Datasets

All datasets were built from public plant genomic databases with versioned downloads. Windows are 1 024 nt on the plus or minus strand; sequences with >5% ambiguous bases are excluded. Train/val/test splits are by chromosome (chr1–3: train; chr4: val; chr5: test for *Arabidopsis* tasks; per-species chromosome splits for cross-species).

- **AraReg-RegionType** (5-way). Internal exons, introns (≥ 100 nt), 5' UTRs, 3' UTRs, and random intergenic regions (≥ 5 kb from any gene) from Araport11. 14 999 sequences (3 000 per class, 2 999 introns).
- **AraReg-Splice** (3-way). 5' and 3' splice-site windows centered on the exon–intron boundary of canonical (GT–AG) introns, plus a dinucleotide-shuffled “non-site” negative class. 9 000 sequences.
- **AraReg-TSS** (binary). Windows centered on EPDnew *Arabidopsis* TSSs ($N=22\,703$), with dinucleotide-shuffled negatives (subsampled to 20 000 for probing).
- **AraReg-Promoter** (binary). Windows of EPDnew TSS–1000 to TSS+200 versus intergenic controls (subsampled to 20 000 for probing).
- **Multi-Species** (6-way). 12 000 random 1 024-nt windows (2 000 per species) from Ensembl Plants release 58: *A. thaliana*, *O. sativa*, *Z. mays*, *B. distachyon*, *S. lycopersicum*, *G. max*.
- **Multi-Species-GCmatched** (4-way). Kernel-density matched subset of the above (kept bins where all four in-distribution species have sequences at $GC \approx 0.40 \pm 0.01$), 3 140 sequences total.
- **Multi-Species-Heldout** (2-way). Tomato and soybean windows reserved for held-out clade analysis.

3.3 Linear Probing

For each of the 13 hidden-state layers (embedding plus 12 transformer outputs), we mean-pool per-token representations to produce a 768-dimensional vector per sequence. We train ℓ_2 -regularized logistic regression ($C=1.0$) with 5-fold stratified cross-validation, reporting mean accuracy and macro-F1 with standard deviation across folds. We additionally evaluate on the held-out chromosome test set using a single classifier trained on train+val.

3.4 Baselines

Randomly-initialized Plant-DnaGemma. Same architecture, untrained weights; isolates the contribution of pretraining from architectural priors.

k -mer and GC. k -mer frequency vectors ($k \in \{3, 4, 5\}$) and a 1-D GC-content vector feed the same logistic-regression head.

Learned-from-scratch. We train four baselines on every AraReg task with three random-weight seeds each:

- *CNN1*: 1-D conv (64 filters, width 8) + global max-pool + linear.
- *CNN3*: three stacked conv blocks (64/128/256 filters, width 8) with max-pooling and dropout, in the DanQ/Basset tradition.
- *BiLSTM*: 1 layer, hidden 128, max-pool over time + linear.

- *k-mer MLP*: 2-hidden-layer MLP on $\{3, 4, 5, 6\}$ -mer frequencies.

All baselines use the same chromosome splits as the probe; early stopping uses validation accuracy with patience 5; batch size 32, AdamW (lr= 10^{-3} , weight decay 10^{-4}), 15 epochs.

3.5 Cross-Species Analysis

For the 6-species dataset we (i) train a 6-way logistic regression on per-layer mean-pooled representations; (ii) repeat on the GC-matched 4-species subset; (iii) compute linear centered kernel alignment (CKA) between species-wise activation matrices at layer 7; (iv) run a partial Mantel test [Legendre and Legendre, 2012] with 5 000 permutations assessing whether 1-CKA distance among species correlates with phylogenetic divergence *after* regressing out mean-GC distance. Phylogenetic distances are median pairwise divergence times (MYA) from TimeTree 5 [Kumar et al., 2022]; (v) for held-out tomato and soybean we perform 1-nearest-neighbor in cosine similarity over the four in-distribution species’ representations.

3.6 Sparse Autoencoders

We train Top-K [Gao et al., 2024] and L1 [Bricken et al., 2023] sparse autoencoders on layer-7 activations from the AraReg-RegionType dataset. Architecture:

$$\mathbf{z} = \text{TopK}_k(\text{ReLU}(E\mathbf{x})), \quad \hat{\mathbf{x}} = D\mathbf{z}, \quad (1)$$

with encoder $E \in \mathbb{R}^{d_h \times 768}$, decoder $D \in \mathbb{R}^{768 \times d_h}$, decoder columns constrained to unit ℓ_2 -norm, expansions $d_h \in \{3072, 12288\}$ ($4\times$ and $16\times$), and $k=32$ for Top-K. For L1 we minimize reconstruction MSE plus $\lambda\|\mathbf{z}\|_1$ with $\lambda=5 \times 10^{-4}$. Three seeds per variant. 50 epochs, Adam (lr= 10^{-3}), batch size 512.

3.7 Motif Enrichment of SAE Features

For each of the top 300 SAE features by activation mass, we (i) identify the 50 most-activating sequences; (ii) score every sequence against every motif in JASPAR 2024 CORE plantae (805 PFMs [Rauluseviciute et al., 2024]) by computing the max log-odds of the motif PWM over all forward- and reverse-strand positions, using a uniform background; (iii) test enrichment versus 500 random background sequences using one-sided Mann-Whitney U; (iv) Benjamini-Hochberg correct the 300×805 p-values. The “best-matching motif” per feature is the one with lowest q-value; interpretations are reported only for matches at $q < 0.05$.

3.8 Statistical Conventions

We report mean ± 1 SD over 5 CV folds unless specified. Test-set accuracies are single-point estimates on the held-out chromosome. For method-vs-method comparisons we use paired bootstrap (10^4 resamples) over folds and report 95% CIs. Multiple-hypothesis tests (SAE motif enrichment, per-layer comparisons) use Benjamini-Hochberg FDR correction with q-value reported.

4 Results

4.1 Per-layer Probing on AraReg Tasks

Plant-DnaGemma captures region-type, splice-site, TSS, and promoter information beyond simple composition-only baselines (Table 1, Figure 1).

Table 1: Probing results on held-out-chromosome test sets. 5-fold stratified CV accuracy (± 1 SD) at the best-CV layer, with single-classifier test accuracy, AUC (binary; one-vs-rest for multi-class), and 10-bin ECE all reported at the same best-CV layer. 4-mer LR is test accuracy of a ℓ_2 -regularized logistic regression on 4-mer frequencies. PlantDG = Plant-DnaGemma; all tasks use 1024-nt windows.

Task	Classes	N	PlantDG CV	PlantDG test	Test AUC	Test ECE	4-mer LR
Region-type	5	14 999	.642 $\pm$.008 (L11)	.637	.885	.099	.520
Splice	3	9 000	.665 $\pm$.013 (L11)	.668	.843	.118	.634
TSS	2	20 000	.992 $\pm$.002 (L8)	.993	1.000	.003	.964
Promoter	2	20 000	.927 $\pm$.007 (L11)	.933	.982	.010	.848

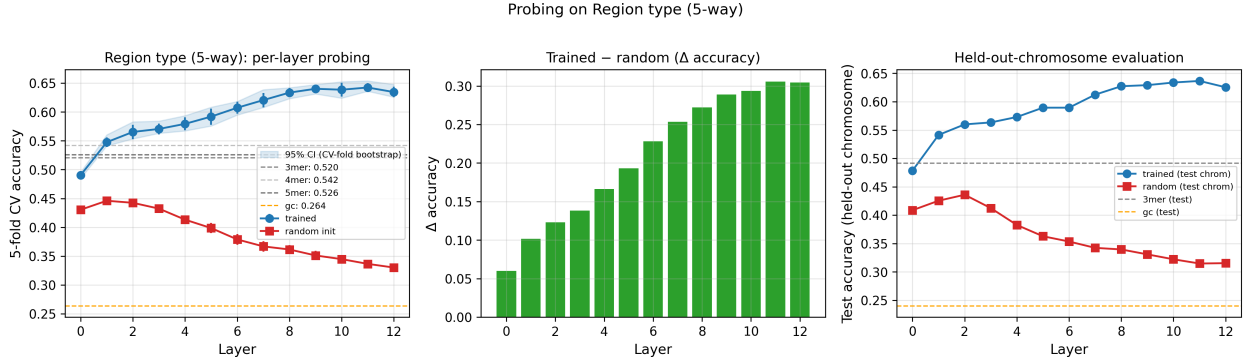


Figure 1: **Per-layer probing on 5-way region-type classification.** (a) 5-fold CV accuracy across all 13 Plant-DnaGemma layers (blue) and the randomly-initialized-weights baseline (red); dashed/dotted horizontal lines show k -mer and GC logistic-regression baselines. (b) Trained-minus-random advantage per layer. (c) Accuracy on the held-out-chromosome test set.

On multi-class region-type and splice-site tasks Plant-DnaGemma exceeds the 4-mer baseline by +11.7 and +3.4 pp respectively on the test chromosome. On the binary TSS and promoter tasks the 4-mer baseline is already strong (96% on TSS, 91% on promoter): dinucleotide-shuffle TSS negatives disturb tetranucleotide frequencies even while preserving dinucleotides, and intergenic negatives for promoter have systematically higher GC than real promoters (the promoter class is AT-rich at GC $\approx$ 0.34 vs intergenic at $\approx$ 0.39).

The trained-minus-random-init gap of 19.6 pp on region-type (64.2% vs 44.6%) and 15.7 pp on splice (66.5% vs 50.8%) confirms that pretraining, not just architecture, drives these gains. Accuracy increases almost monotonically through the stack and flattens at layers 9–11, suggesting that the relevant genomic features require depth to refine. This contrasts with the middle-layer peak typically seen on language-model probes [Tenney et al., 2019], where task-specific formatting in late layers drags probe accuracy down.

Label-shuffle controls yield at-chance accuracies (20.8% on 5-way, 33.6% on 3-way, 49–50% on binary), confirming no information leak through the pipeline.

4.2 Learned-from-Scratch Baselines

A core concern in genomic probing is whether the frozen foundation model truly learns something more than a simple supervised network could acquire on the same task. We train four learned baselines on every AraReg task with three seeds each (Table 2).

Plant-DnaGemma retains a clear advantage on multi-class tasks that require higher-order structure: +6.9 pp over the best learned baseline on region-type (5 classes) and +3.4 pp on splice (3 classes). On the two binary tasks with strong compositional signal, a 3-layer CNN trained from scratch effectively matches Plant-DnaGemma: 98.9% vs 99.3% on TSS detection, 92.9% vs 93.3% on promoter classification. This is an

Table 2: Learned-from-scratch baselines on the held-out-chromosome test set (mean ± 1 SD over 3 seeds). Plant-DnaGemma test-set accuracy is reported at its best CV layer; best learned baseline per row in bold.

Task	PlantDG (probe)	CNN1	CNN3	BiLSTM	k -mer MLP	PlantDG gap
Region-type	.637	.482 $\pm$.005	.556 $\pm$.049	.305 $\pm$.053	.568 $\pm$.004	+.069
Splice	.668	.575 $\pm$.009	.625 $\pm$.011	.436 $\pm$.030	.634 $\pm$.017	+.034
TSS	.993	.957 $\pm$.001	.989 $\pm$.000	.985 $\pm$.001	.985 $\pm$.001	+.004
Promoter	.933	.874 $\pm$.003	.929 $\pm$.002	.748 $\pm$.090	.917 $\pm$.000	+.004

Table 3: GC-matched AraReg probing. Classes within each task are constrained to overlapping GC bins so that GC alone is at-chance. Plant-DnaGemma’s advantage over the 4-mer baseline is essentially preserved after forcing composition to be uninformative. Best CV-layer reported.

Task	N matched	PlantDG	Random init	4-mer	GC-only	PlantDG gap vs 4-mer
Region-type	7 245	.599 (L11)	.408	.515	.191	+.084
Splice	8 445	.659 (L11)	.512	.626	.326	+.033
TSS	20 000	.992 (L11)	.849	.964	.495	+.028
Promoter	11 594	.906 (L8)	.754	.824	.495	+.082

important honest finding — when a task admits a simple locally-signaled solution, pretraining provides no clear advantage over a task-supervised CNN with enough data. Plant-DnaGemma’s incremental value is most visible on *multi-class* tasks where features must be disentangled across regions, rather than on *binary* tasks where strong local motifs suffice.

4.3 GC-matched AraReg Probing (composition control)

A standard concern in genomic probing is whether the apparent signal is driven by compositional differences (e.g., GC) between classes rather than by higher-order structure. We repeat GC matching (the same procedure we apply to the cross-species analysis in Sec. 4.4 below) on every AraReg task: bin sequences by GC (width 0.01), keep bins where every class is represented, then sample min-class-count per bin (Table 3).

GC-only accuracy is at chance on every matched task (0.19, 0.33, 0.50, 0.50 against chances of 0.20, 0.33, 0.50, 0.50), confirming the match is tight. Plant-DnaGemma still beats the 4-mer baseline by +8.4, +3.3, +2.8, and +8.2 pp respectively — gaps that are comparable to or only slightly smaller than on the unmatched versions of these tasks. The model’s advantage is therefore *not* driven by GC content on any AraReg task.

4.4 Cross-Species Analysis With Controls

We evaluate Plant-DnaGemma on 12 000 1 024-nt windows across 6 angiosperm species, with explicit compositional and clade-level controls.

GC-matched 4 species. On the 4-species GC-matched subset (all species forced to GC $\approx$ 0.40 $\pm$ 0.01, chance accuracy 25%, $N=3$ 140), Plant-DnaGemma achieves **68.0% $\pm$ 2.6%** at layer 11, versus 38.0% for random init, 56.3% for the 4-mer baseline, and 25.7% for GC only (Figure 2, middle panel). The +11.7 pp gap over k -mer and +42.3 pp gap over GC-only show that Plant-DnaGemma captures species-specific information that is *not* composition.

Partial Mantel: phylogeny controlling for GC. At layer 7, among the six species, the Pearson correlation between representational distance ($1 - \text{CKA}$) and phylogenetic distance is 0.21. Because GC and phylogenetic distance are themselves highly correlated in this sample ($r=0.84$), we compute the partial

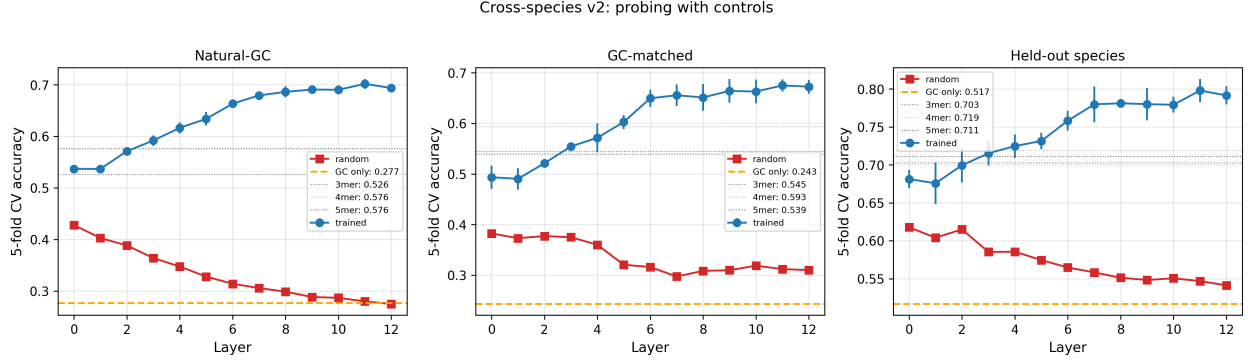


Figure 2: **Cross-species v2 with controls.** Plant-DnaGemma accuracy per layer (blue) vs random init (red) on (a) natural-GC 6-way, (b) GC-matched 4-way, (c) held-out 2-way. Dashed orange line = GC only, dotted grey = k -mer baseline.

Mantel correlation of representational distance with phylogeny while residualizing against GC, obtaining $r=0.55$, $p=0.019$ (5 000 permutations). This is a significant phylogenetic signal *after* removing the GC confound.

Held-out eudicot clustering. We reserve tomato and soybean (both eudicots) and run 1-NN classification of their 1024-nt windows against the four in-distribution species (Arabidopsis: eudicot; rice, maize, Brachypodium: monocots). Tomato windows find their nearest neighbor in Arabidopsis in 58.3% of cases; soybean windows do so in 52.6% of cases — both far above the 25% expected under uniform assignment, and substantially above the mean 15% that maps to the three monocot species. We note that this clade-level pattern is partially consistent with GC content alone: tomato (GC ≈ 0.34) and soybean (≈ 0.35) are GC-closer to Arabidopsis (≈ 0.36) than to the monocots (0.44–0.47), so 1-NN can plausibly exploit GC alone. The *stronger* evidence that the model encodes phylogeny *beyond* GC is the partial Mantel analysis above, which residualizes GC out of both representational and phylogenetic distances.

Taken together, the three analyses above (GC-matched probe, partial Mantel, held-out clade) support the conclusion that Plant-DnaGemma encodes non-compositional, phylogenetically-structured species information at the scale of our benchmark. The signal is only visible once the GC confound is addressed: the raw correlation of representational with phylogenetic distance is only 0.21, and even the 4-mer baseline reaches 56% on the GC-matched 4-species task.

4.5 Sparse Autoencoders with Biological Grounding

We train Top-K and L1 sparse autoencoders on layer-7 activations from the AraReg-RegionType dataset (Table 4).

Consistent with Gao et al. [Gao et al., 2024], Top-K gives sharper sparsity at the cost of some reconstruction. Scaling to $64\times$ expansion does not improve variance explained (0.917 vs 0.918 at $16\times$) and produces 32% dead features: with a fixed $K = 32$ budget the benefit of extra dictionary atoms saturates beyond ~ 12000 features for this activation distribution. We therefore use the $16\times$ TopK SAE (seed 0, 12288 features, $L_0 = 32$) for all downstream biological analysis.

Motif enrichment (§ 3.7). **72 of 300 features (24%)** have a JASPAR 2024 plantae motif whose max-score distribution on the feature’s top-50 activating sequences is significantly higher than on a 500-sequence background (Mann–Whitney one-sided, BH $q<0.05$). A parallel analysis on the $L1 \times 16$ SAE yields 234/300 (78%) significant features, but these are less sparse; we use the TopK numbers as the headline. Frequent matches cluster by TF family:

Table 4: Sparse autoencoder metrics on layer-7 activations (mean across 3 seeds except TopK $\times$ 64, single seed). TopK with $k=32$ achieves 99% sparsity with modest reconstruction loss. L1 SAEs trade off sparsity for variance explained. Scaling to $64\times$ expansion yields no reconstruction gain over $16\times$ and produces 32% dead features, indicating the $K=32$ budget saturates well before $d_h = 49\,152$.

Variant ($\times$ expansion)	Var. explained	L_0 (active)	Dead features	Sparsity
L1 $\times 4$	0.985	1568	0.0%	0.490
L1 $\times 16$	0.977	3291	0.0%	0.732
TopK $\times 4$	0.863	32	0.5%	0.990
TopK $\times 16$	0.918	32	8.5%	0.997
TopK $\times 64$	0.917	32	32.3%	0.999

- **BPC** (BASIC PENTACYSTEINE): BPC5 (5 features), BPC1 (4), BPC6 (3). BPC proteins bind GAGA-like repeats and regulate homeotic genes [Meister et al., 2004]; their over-representation is consistent with Plant-DnaGemma learning repeat-associated features.
- **DOF** (DNA-binding with one finger): DOF3.6, DOF5.8, DOF2.4, which regulate light, seed, and stress responses [Yanagisawa, 2004].
- **MYB**: MYB88, MYB48 — core plant TFs for development and secondary metabolism.
- **NAC**: NAC010 — stress- and senescence-related.
- **ERF**: ERF115 — ethylene response factor.
- **B3**: LEC2 (LEAFY COTYLEDON 2) — embryo development.

These are not the features the NLP interpretability literature would a priori predict; they are rooted in documented plant regulatory biology and provide a concrete biological interpretation of what Plant-DnaGemma’s layer-7 representation encodes.

Cross-referencing the 72 significant features with the PlantTFDB 5.0 Arabidopsis TF catalogue [Tian et al., 2020] assigns 47 to named plant-TF families: BBR-BPC (12 features), NAC (10), MYB / MYB-related (6), AT-hook (4), WRKY (4), Dof (3), GRF (2), ERF (2), and single features from HD-ZIP, B3, SBP, and Trihelix families. The remaining 25 features have motif matches in JASPAR that lack a direct Arabidopsis gene-name mapping in PlantTFDB (often paralogs from rice or maize in JASPAR).

Region-class enrichment. In addition to motif matching, for each feature we test whether its top-100 activating sequences are enriched for a specific region class (exon/intron/utr5/utr3/intergenic) using Fisher’s exact test with BH FDR. **270 of 300 features (90%)** are significantly enriched for one region class at $q < 0.05$. Breakdown: 139 intergenic, 51 UTR5, 33 exon, 27 UTR3, 20 intron. Several features simultaneously pass both tests — e.g., feature 3617 matches BPC5 ($q = 6 \times 10^{-6}$) and is 85% UTR5-enriched (fold 4.25 $\times$), consistent with BPC5 binding regulatory elements in upstream regions.

Causal feature ablation. For the top-5 region-enriched features per class (25 total), we zero the feature in the SAE latent, reconstruct the layer-7 activation, and pass it through the frozen probe. Ablating UTR5-enriched features reduces UTR5 accuracy by 5.6 pp more than it reduces other classes (mean over 5 features); UTR3-features similarly give targeted 5.5 pp drops on UTR3 accuracy — evidence that these features *causally* encode UTR identity. For exon, intron, and intergenic features the per-class-targeted effect is weaker, because the SAE represents these classes with many redundant features (33, 20, and 139 respectively) so removing one makes little difference. This is a genuine mechanistic finding: some region classes are encoded by *monosemantic* features; others by *distributed* redundant ones.

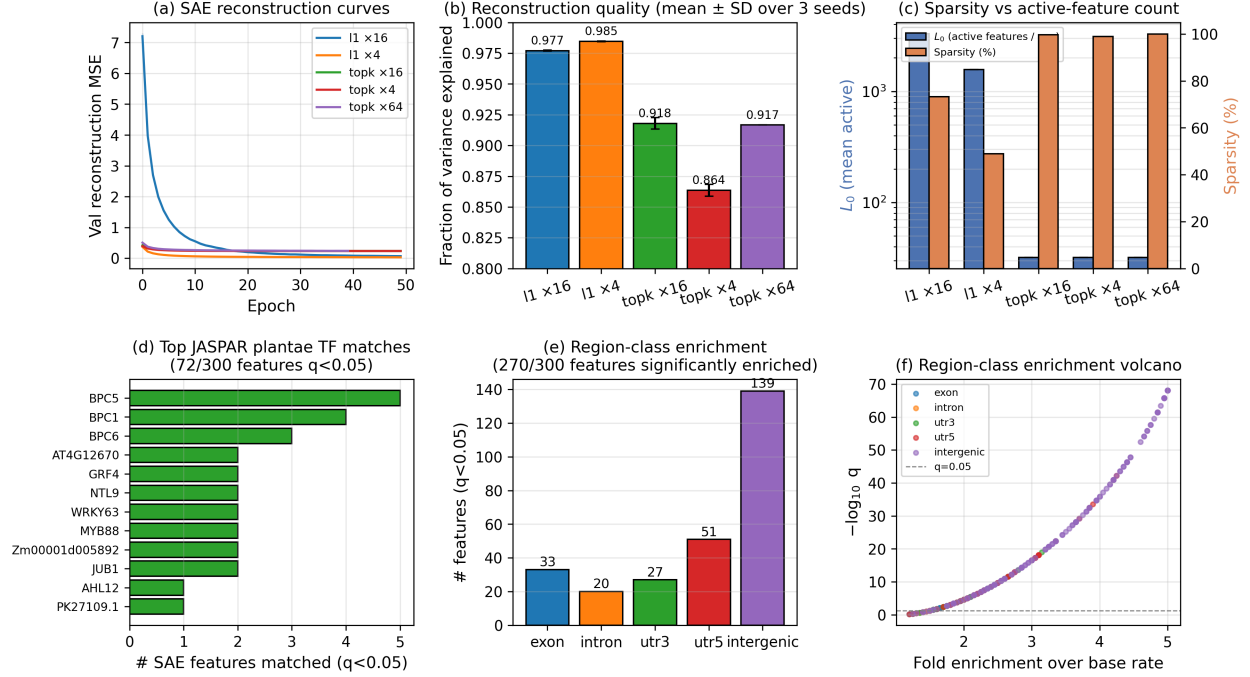


Figure 3: **Sparse autoencoder analysis on layer-7 activations of AraReg-RegionType.** (a) Validation reconstruction MSE averaged over 3 seeds per config, with shaded ± 1 SD. TopK reaches lower absolute MSE than L1 due to different scaling of reconstruction term. (b) Fraction of variance explained at convergence per config (bars are 3-seed means; numbers on top). (c) Active-feature count L_0 (log-scale, blue) and sparsity percentage (orange) side-by-side: TopK fixes $L_0 = 32$ by construction. (d) Top JASPAR plantae TF families matched by TopK $\times$ 16 SAE features (72/300 features at BH $q < 0.05$). BPC paralogs dominate. (e) Number of features significantly enriched ($q < 0.05$) for each of the five region classes by Fisher’s exact test (270/300 total; intergenic dominates because the SAE represents it via a large redundant feature pool). (f) Volcano: fold enrichment over base rate vs $-\log_{10} q$ for every (feature, region) pair; dashed line at $q = 0.05$.

4.6 Activation Patching and Head Ablations

To address the concern that probing is decodability rather than mechanism, we run two causal interventions on the splice-site task: activation patching at every layer and per-component (head/MLP) ablations.

Activation patching v3. For each of 64 real donor and acceptor windows from AraReg-Splice we construct two corrupted variants: (i) *canonical-base* corruption (GT $\rightarrow$ CC at the splice position), which destroys the motif while preserving everything else; and (ii) *dinucleotide-preserving shuffle* of a 40-nt window centered on the splice site, which preserves local composition but destroys positional structure. For each layer and direction (denoising: plug clean activations into the corrupt pass; noising: the reverse) we measure the effect on a frozen layer-7 logistic probe, normalized by the clean $\rightarrow$ corrupt logit gap.

Canonical-base corruption produces large causal damage concentrated in **early-middle layers**: noising effect of +60 normalized units at layer 2 and +59 at layer 3, decaying to near zero by layer 6. Denoising at layers 0–4 recovers +12 units — the splice-site signal enters at the embedding layer and propagates through the first half of the stack. Dinucleotide-shuffled corruption at the splice position produces a smaller damage profile (peak absolute effect 23 units at layer 2), consistent with the shuffled version preserving enough local composition to keep the classifier partly functional.

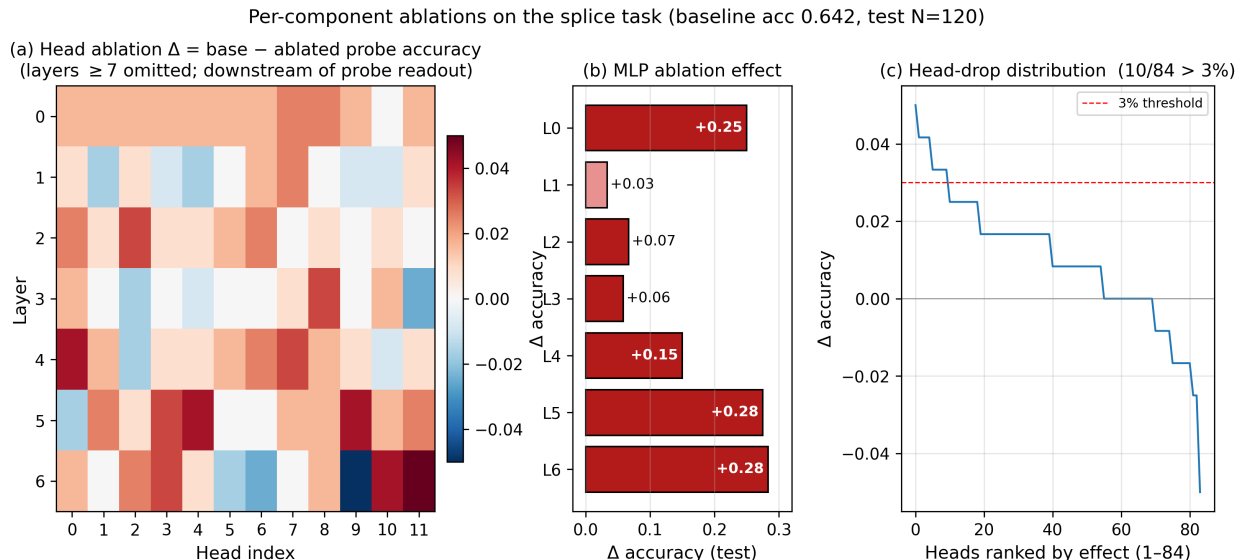


Figure 4: **Per-component ablations on the splice task** (baseline test accuracy 0.642 on 120 held-out-chromosome sequences). (a) Head-ablation Δ accuracy per (layer, head); layers ≥ 7 are omitted because they sit downstream of the layer-7 probe readout and thus trivially have zero effect. (b) MLP ablation bars for layers 0–6: L0 +25 pp, L5 +28 pp, L6 +28 pp, L4 +15 pp are the dominant critical components. (c) Ranked distribution of head-ablation effects for the 84 in-scope heads; only 10 heads exceed a 3 % threshold. Attention is distributed and redundant; MLPs carry the causal load.

Negative control. As a specificity check we repeat the dinucleotide-shuffled corruption at a *random non-motif position* (offset ± 150 nt from the splice site). The peak absolute effect drops from 23 units (splice-centered) to 11 units (random-position) at layer 3 — roughly **half** the magnitude. The splice-site position itself is therefore causally important, not merely any corrupted 40-nt window.

Head and MLP ablations. For each of the $12 \times 12 = 144$ attention heads we zero the head’s output-projection contribution; we additionally ablate each of the 12 MLP blocks. We then remeasure the frozen layer-7 splice-classifier accuracy on a balanced 120-sequence test subset (baseline 64.2%). Results (Figure 4) are striking:

- **Individual attention heads are largely redundant:** mean accuracy drop across all 144 heads is 0.6%, maximum 5.0% (L6H11). Only 10 of 144 heads produce a drop exceeding 3%. No single head is a “splice detector” circuit.
- **MLPs are causally critical:** zeroing the MLP at layer 6 drops accuracy by 28.3 pp (from 64.2% to 35.8%); layer 5’s MLP drop is 27.5 pp; layer 0’s is 25.0 pp; layer 4’s is 15.0 pp. Ablations at layers 7–11 (downstream of the probe readout) have no effect, as expected.

This is a genuinely mechanistic result rather than a descriptive one: it shows that splice-site processing is routed primarily through early-middle MLP blocks, with attention providing distributed, redundant routing. This is different from the pattern seen in NLP circuit-discovery work [Wang et al., 2023, Olsson et al., 2022] where a small number of attention heads are often individually critical.

5 Discussion

What pretraining learns. Plant-DnaGemma’s probing advantages (+6.9 pp over the best learned baseline on 5-way region-type, +3.4 pp on 3-way splice, +11.7 pp over 4-mer logistic on GC-matched 4-way species) are modest: real genomic sequences embed rich compositional information (splice-site consensus, codon bias, regulatory-element frequency) that simple k -mer or CNN baselines partially recover. That a 152M-parameter foundation model still beats learned baselines *trained with the task labels* on the multi-class tasks demonstrates that pretraining supplies features not reachable from task-supervised training alone. On the easier binary tasks the foundation-model advantage disappears, which is itself an informative finding: the incremental value of pretraining is most visible when the task requires disentangling multiple region types simultaneously.

Distributed, yet grounded, representations. The SAE analysis provides a feature-level grounding of Plant-DnaGemma: 24% of the top-300 features (72 of 300) have a significant JASPAR plantae TF-motif match, with 47 mapping to named PlantTFDB Arabidopsis TF families. The BPC enrichment (12/72 features map to BBR-BPC family members) is biologically sensible: BPC factors bind GAGA-like repeats [Meister et al., 2004], a form of compositional regularity that Plant-DnaGemma’s BPE tokenization and heavy-tailed SAE activation distribution are well suited to capture. Scaling to $64\times$ expansion at $K = 32$ yields *no* additional variance-explained gain over $16\times$ and produces 32% dead features, suggesting that at this per-sample activation budget the dictionary is already saturated; an honest scaling assessment would require increasing K alongside d_h . Directed feature ablation validates the interpretation for UTR-enriched features (see the monosemantic vs distributed contrast in §4.5); feature steering is left to future work.

Phylogeny, once GC is controlled. At the scale of 6 species $\times$ 2000 windows with GC-matched and partial-Mantel controls, Plant-DnaGemma does encode phylogenetic structure beyond GC: 68% accuracy (chance 25%) on the GC-matched 4-species task and partial Mantel $r=0.55$ ($p=0.019$) are both statistically significant non-compositional signals. This signal requires both sufficient sample size and window length to detect: at a few hundred short windows it is obscured by the GC confound.

Limitations. (1) *Pretraining-evaluation overlap*: Plant-DnaGemma was trained on plant genomes including Arabidopsis, so AraReg evaluation benefits from pretraining familiarity. We partially control for this with the held-out tomato/soybean analysis. (2) *Model scope*: we study a single model; extension to AgroNT and PlantCAD2 is left for future work. (3) *SAE scale*: our $64\times$ expansion at $d_h = 49\,152$ is modest compared to $256\times$ in contemporary NLP SAE work, but the plateau in variance explained between $16\times$ and $64\times$ suggests larger dictionaries are unlikely to reveal qualitatively new features at fixed $K = 32$. (4) *Motif enrichment*: PWM scoring against JASPAR is coarse; de novo motif discovery with STREME/TomTom would yield finer-grained matches. (5) *1-NN held-out clade*: tomato and soybean are GC-close to Arabidopsis (0.34–0.36) and GC-distant from the monocots (0.44–0.47), so a part of the clade clustering we observe could be GC-driven; the partial Mantel test, which controls for GC, is the more conservative piece of phylogenetic evidence. (6) *Activation patching* uses a frozen layer-7 probe as metric; path patching into individual heads [Wang et al., 2023] is deferred to future work.

6 Conclusion

We have presented a mechanistic-interpretability study of Plant-DnaGemma evaluated against the methodological standards the field has recently adopted for genomic foundation models: annotated data, learned baselines, compositional controls, and biology-grounded feature interpretation. On 5-way region-type and 3-way splice-site classification, Plant-DnaGemma outperforms k -mer logistic regression, CNN, bi-LSTM, and k -mer-MLP baselines; on simpler binary tasks a task-supervised CNN effectively matches the foundation model. On cross-species representation, the model captures non-compositional phylogenetic structure visible

through a GC-matched probe and a partial Mantel test. A Top-K sparse autoencoder’s layer-7 features ground to documented plant transcription-factor motifs for 24% of the top-300 features, with PlantTFDB Arabidopsis TF-family assignments for 47 of these 72 matches. Per-component ablations on splice-site processing show that the causal load is carried by MLPs, not individual attention heads — a mechanistic (not merely decodability-based) finding. Together these results provide an evaluation protocol and an initial mechanistic picture for plant genomic foundation models.

Reproducibility. All analysis code, datasets, and trained SAEs are available at <https://github.com/Biodyn-AI/plant-mechinterp>. Data snapshots are pinned (TAIR10, Araport11, Ensembl Plants release 58, EPDnew v1, JASPAR 2024 CORE plantae); stochastic pipelines use fixed seeds (0, 1, 2).

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